Four further structural tables over $\mathbb{F}_4$

This finite application proves the displayed subgroup descriptions in four current Connor–Leemans companion tables. Complete numerical inventories, the exact source bijections and the actual 44-point models are acquired from the accepted nine-table restriction component. No ambient subgroup census is repeated.

Exact assertion and source convention. Relative to the acquired named atoms and earlier certified registry, each of the following 538 actual embedded subgroups admits the description in its source row. A colon asserts a split extension, a dot a nonsplit extension, and a cross a direct product. These descriptions need not determine a unique abstract group. Every occurrence is verified separately.

Current file	Displayed top group	Rows	Pages
`psl34d21`	$L_3(4) : 2$	151	6
`psl34d23`	$L_3(4) : 2$	102	4
`psl34d6`	$L_3(4) : 6$	143	6
`psl34ds3`	$L_3(4) : S_3$	142	6
Total		538	22

All twenty-two native pages were directly inspected, including the final continued-table page without data in `psl34d6`. Powers, punctuation and the default left association are recorded literally. In particular, `psl34d6` row 57 is $Q_8 \cdot 6$, whereas rows 51 and 56 are $Q_8 : 6$. The nested description in `psl34ds3` row 43 is $(D_8 \cdot 2^2) : 2$. Products do not erase an earlier dot: `psl34d21` row 19 is $(((3 : S_3) \cdot 2) \times 2) \cdot 2$.

The actual models are the accepted subgroups of the 44-point $\mathrm{Aut}(L_3(4))$ model: field times inverse-transpose, inverse-transpose, diagonal together with field times inverse-transpose, and diagonal together with inverse-transpose, respectively. Their outer groups are C_2, C_2, C_6, S_3. These are exact acquired constructions; the filename suffixes are not asserted to define a canonical numbering of outer involutions. The two printed $L_3(4) : 2$ names are therefore not silently identified. Each full and proper subgroup is parsed from literal permutation generators in its own saved state; display-name constructors cannot replace its embedding.

The named $L_3(4)$ atom is the previously certified projective model, acquired through the M22/L100 interfaces and the preceding 347-row application. There are 161 acquired registry entries and no new named simple or Mathieu atom. Candidate discovery adds 37 small structural models: 29 split, five nonsplit, two direct and one extraspecial group. The saved generator images, rather than an isomorphism-search answer or group-order comparison, supply each new identification.

Complete certificate argument. We acquire the labelled-generator transport criterion. A bijection from a finite model to an actual subgroup, fixing identity and preserving right multiplication by every ordered generator, is a homomorphism by induction on words. Hence it is an isomorphism. The reader enumerates both actual finite groups by permutation multiplication, verifies the entire bijection and checks every generator edge. References to certified factors are processed in dependency order.

For a split description the certificate supplies $N, K \leq H$ with their acquired or previously checked names. It checks normality of N on the generators of H, $N \cap K = 1$, and $|N||K| = |H|$. Thus $H = NK$ and the description follows. A noncommuting generator pair confirms the printed nontrivial action. For a direct product the factor generators commute, so the same generation and intersection checks give $H = N \times K$.

Three new nonsplit models have quotient C_2. Every element outside the specified normal subgroup is checked to have square different from identity. Any complement would have order two and contradict this check. The other two have quotient C_2^2. Four literal cosets, with the quotient relations, establish this quotient. All lifts of its two fixed basis elements are tested: no pair consists of commuting involutions. A complement would lift that basis to such a pair, so no complement exists. The printed $Q_8 \cdot 6$ description uses an earlier accepted model and its cyclic-quotient lift obstruction; it is not inferred from the absence of a found complement.

For the new group of order 32, literal multiplication shows that the center and derived subgroup are the same group of order two and that every square lies there. Thus the quotient by the center is elementary abelian and the Frattini subgroup is that same center: the group is extraspecial. The source symbol 2^{1+4} makes no further sign choice. All these arguments and the finite readers are acquired criteria applied to the new explicit arrays, not new classification premises.

Before any structural certificate is considered, the reader checks the source row, order, syntax tree, accepted source-to-computation bijection, actual state generators and degree. Each top-row extension is checked by the same literal certificates as every proper row. All 538 occurrences are required; reuse of an abstract model never skips its embedded join.

Evidence and limits. One bounded witness pass returned all 538 proposals in 11.58 seconds, with clean output and no missing row. Independent literal verification uses only the retained arrays. Ordinary and optimized Python must give identical reports. Damaged controls test source association, powers, dots, embedded generators, exact degrees, literal-state grammar and completion markers. Small quaternion examples separately test the complete lift obstructions for Klein-four and cyclic-six quotients.

The current companion PDFs credit Thomas Connor and Dimitri Leemans and print November 29, 2012. Their exact present bytes are bound; identity with a particular historical 2013 or 2015 file is not asserted. Existing numerical completeness and named-constructor interfaces remain explicit inputs. No ambient census, unique extension-name classification, novelty, whole-original, conjecture, formalization or measurement credit is claimed.